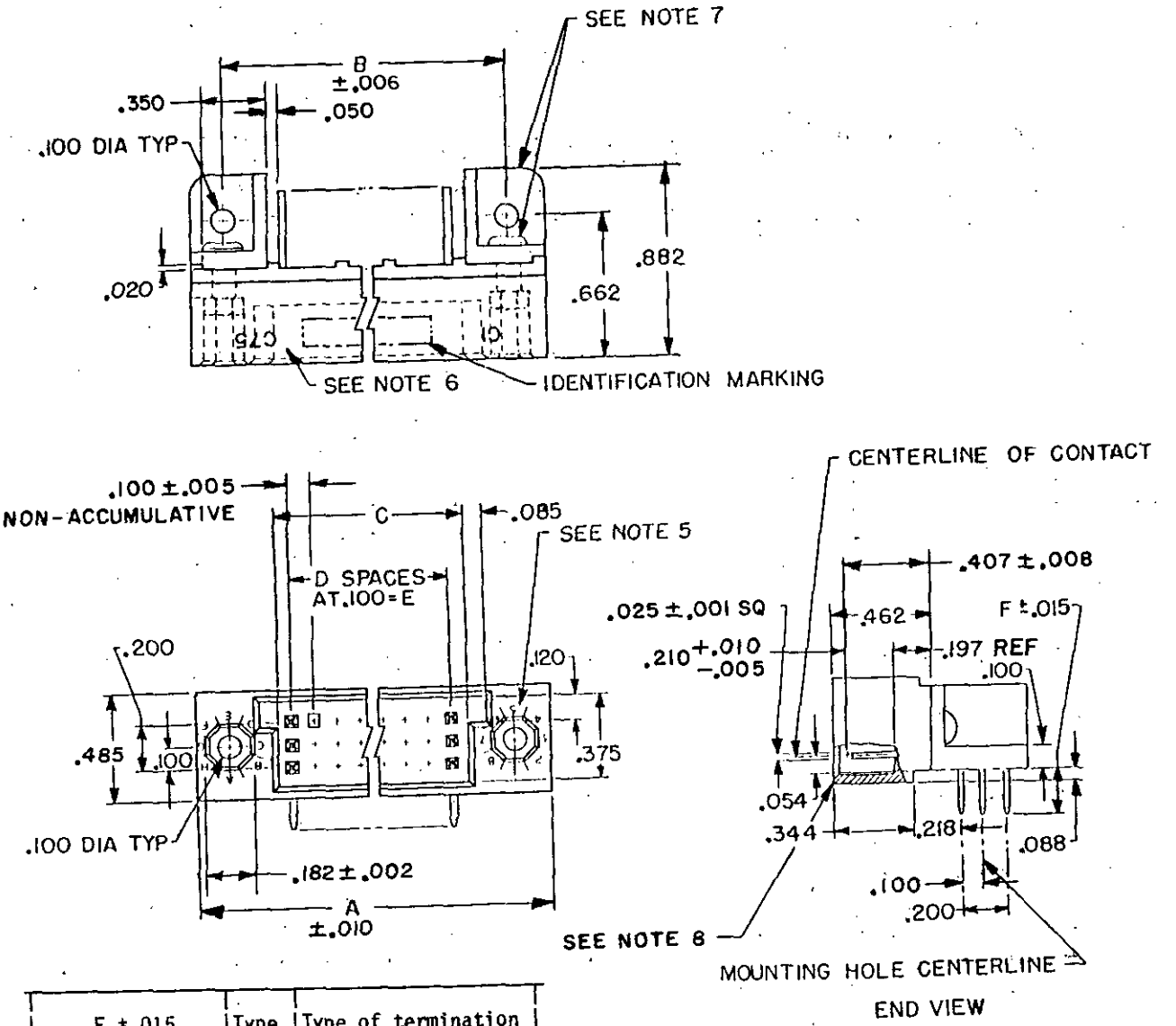
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 3-ROW PIN ASSEMBLIES, RIGHT ANGLE, 75 THROUGH 405 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



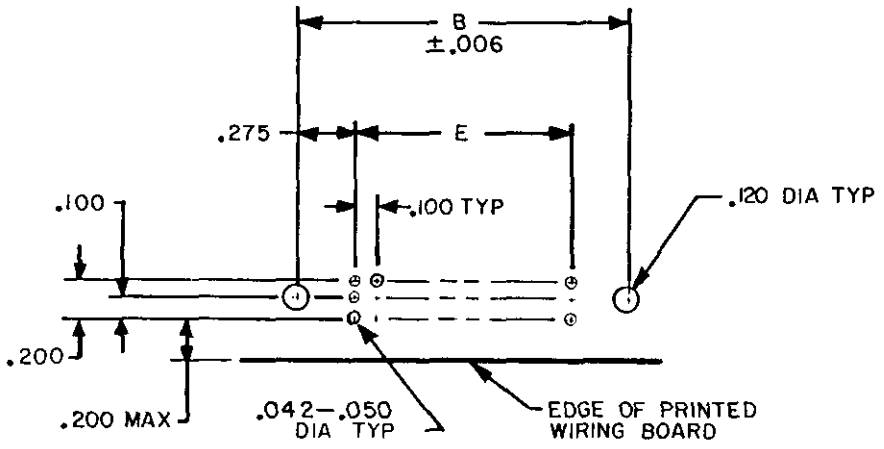
F ±.015		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post
.165	4.19	II	Solder post
.140	3.56	III	Solder post
.120	3.05	IV	Solder post

FIGURE 1. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing, 75 through 225 contacts.

AMSC N/A

FSC 5935

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.C15	0.38
.025	0.64
.042	1.07
.050	1.27
.054	1.37
.085	2.16
.088	2.24
.100	2.54
.120	3.05
.182	4.62
.197	5.00
.200	5.08
.210	5.33
.218	5.54
.275	6.98
.334	8.48
.350	8.89
.375	9.52
.407	10.34
.462	11.73
.485	12.32
.662	16.81
.882	22.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/177.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and markings every ten positions stamped this surface.
7. Mounting ears and hardware are purchased separately. See MIL-C-55302/182 for mounting hardware.
8. Chamfers shall be .015 minimum at a 45° angle ±5 degrees (see figure 1, 2, or 3).

FIGURE 1. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing, 75 through 225 contacts - Continued.

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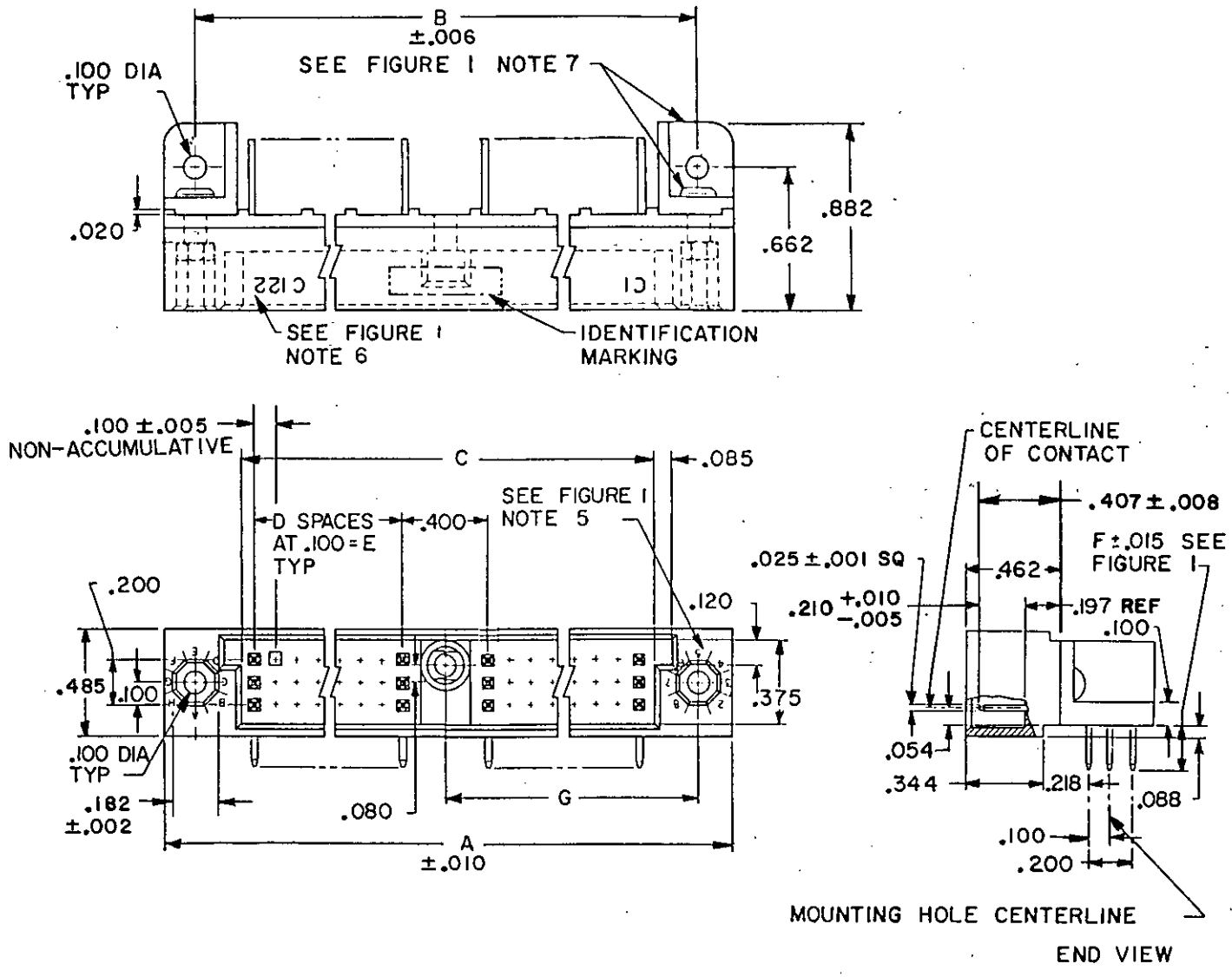
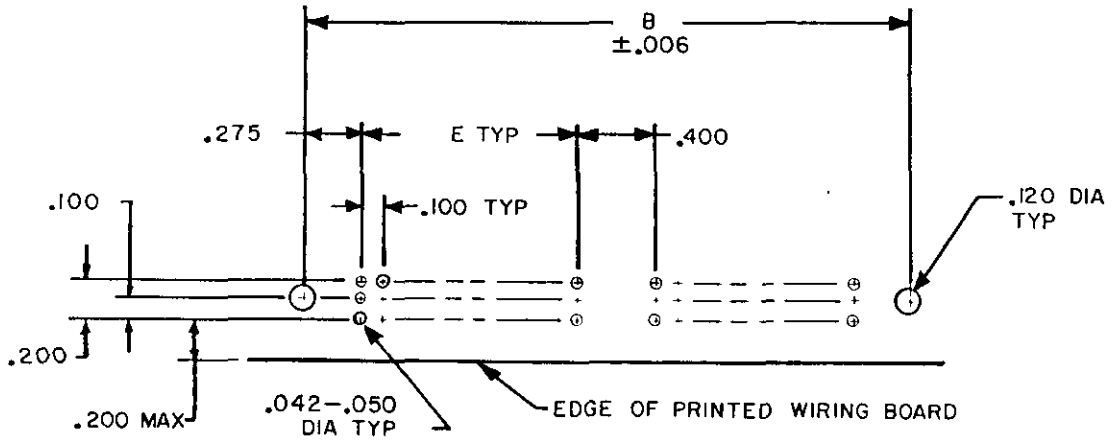


FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide, 240 through 366 contacts.

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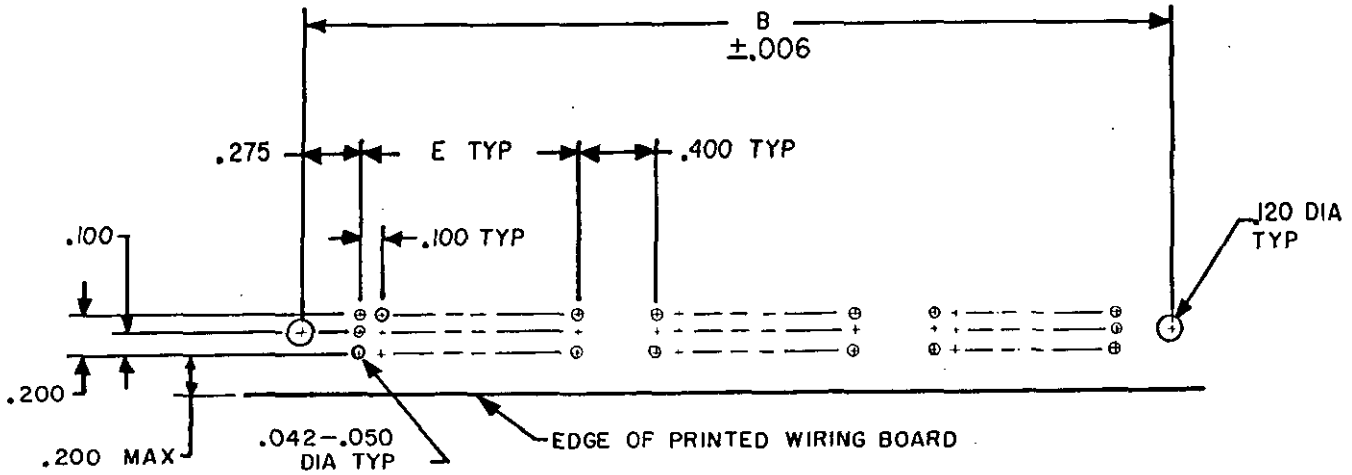
Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.042	1.07
.050	1.27
.080	2.03
.085	2.16
.088	2.24
.100	2.54
.120	3.05
.182	4.62
.197	5.00
.200	5.08
.210	5.33
.218	5.54
.275	6.98
.310	7.87
.344	8.74
.400	10.16
.407	10.34
.462	11.73
.485	12.32
.662	16.81
.882	22.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide, 240 through 366 contacts - Continued.

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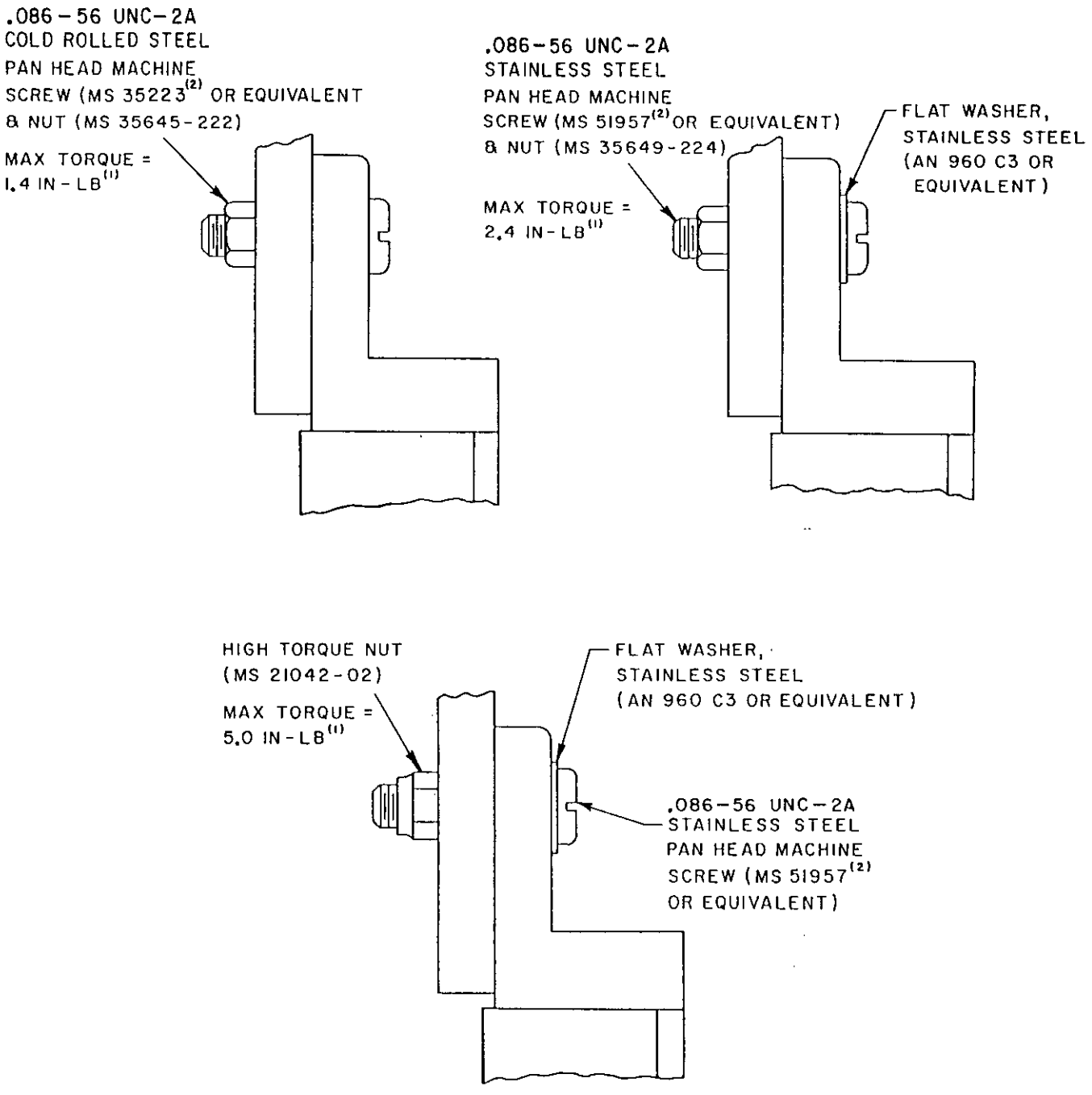
Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.042	1.07
.050	1.27
.080	2.03
.085	2.16
.088	2.24
.100	2.54
.120	3.05
.182	4.62
.197	5.00
.200	5.08
.210	5.33
.275	6.98
.344	8.74
.375	9.52
.400	10.16
.407	10.34
.462	11.73
.480	12.19
.662	16.81
.882	22.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with two guides, 387 through 405 contacts - Continued.

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NOTES:
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 4. Mounting recommendations.

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 03 04	75	1	I II III IV	3.250 (82.55)	2.950 (74.93)	2.524 (64.11)	24	2.400 (60.96)			1-25
05 06 07 08	90		I II III IV	3.750 (95.25)	3.450 (87.63)	3.024 (76.81)	29	2.900 (73.66)			1-30
09 10 11 12	96	1	I II III IV	3.950 (100.33)	3.650 (92.71)	3.224 (81.89)	31	3.100 (76.74)			1-32
13 14 15 16	105	1	I II III IV	4.250 (107.95)	3.950 (100.33)	3.524 (89.51)	34	3.400 (86.36)			1-35
17 18 19 20	120	1	I II III IV	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)			1-40
21 22 23 24	135	1	I II III IV	5.250 (133.35)	4.950 (125.73)	4.524 (114.91)	44	4.400 (111.76)			1-45
25 26 27 28	150	1	I II III IV	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)			1-50
29 30 31 32	165	1	I II III IV	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)			1-55
33 34 35 36	180	1	I II III IV	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)			1-60
37 38 39 40	195	1	I II III IV	7.250 (184.15)	6.950 (176.53)	6.524 (165.71)	64	6.400 (162.56)			1-65

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
41 42 43 44	201	1	I II III IV	7.450 (189.23)	7.150 (181.61)	6.724 (170.79)	66	6.600 (167.64)			1-67
45 46 47 48	210	1	I II III IV	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)			1-70
49 50 51 52	225	1	I II III IV	8.250 (209.55)	7.950 (201.93)	7.524 (191.11)	74	7.400 (187.96)			1-75
53 54 55 56	240	2	I II III IV	9.050 (229.87)	8.750 (222.25)	8.324 (211.43)	39	3.900 (99.06)	4.375 (111.13)		1-80
57 58 59 60	258	2	I II III IV	9.650 (245.11)	9.350 (237.49)	8.924 (226.67)	42	4.200 (106.68)	4.675 (118.75)		1-86
61 62 63 64	276	2	I II III IV	10.250 (260.35)	9.950 (252.73)	9.524 (241.91)	45	4.500 (114.30)	4.975 (126.37)		1-92
65 66 67 68	294	2	I II III IV	10.850 (275.59)	10.550 (267.97)	10.124 (257.15)	48	4.800 (121.92)	5.275 (133.99)		1-98
69 70 71 72	312	2	I II III IV	11.450 (290.83)	11.150 (283.21)	10.724 (272.39)	51	5.100 (129.54)	5.575 (141.61)		1-104
73 74 75 76	330	2	I II III IV	12.050 (306.07)	11.750 (298.45)	11.324 (287.63)	54	5.400 (137.16)	5.875 (149.23)		1-110

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
77 78 79 80	348	2	I II III IV	12.650 (321.31)	12.350 (313.69)	11.924 (302.87)	57	5.700 (144.78)	6.175 (156.85)		1-116
81 82 83 84	366	2	I II III IV	13.250 (336.55)	12.950 (328.93)	12.524 (318.11)	60	6.000 (152.40)	6.475 (164.47)		1-222
85 86 87 88	387	3	I II III IV	14.250 (361.95)	13.950 (354.33)	13.524 (343.51)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
89 90 91 92	405	3	I II III IV	14.850 (377.19)	14.550 (369.57)	14.124 (358.75)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135

TABLE II. Contact plating requirements.

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area shall have a minimum plating in accordance with MIL-C-55302. Solder post area (contact type I, .195 minimum length; type II, .180 minimum length; type III, .155 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (86/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290, and shall meet the requirements of MIL-P-45209.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, 3, 4, and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .094; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number: M55302/178 (and dash number from table I and type from table II).

Example: M55302/178-01A.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5935-3542-06)

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